

#UNIT-1

#Computer Vision —

(i) Extract meaningful info from images/videos

(ii) Mimic human vision

(iii) Convert raw pixel data into high level understanding

(iv) Applications: Autonomous vehicles, medical imaging

(v) Challenges: Noise, illumination changes

#Vision System Architecture —

Pipeline:

(i) Image acquisition (camera)

(ii) Analog to Digital Conversion

(iii) Image Processing

(iv) Decision/Output

#Optical System (Lens & Aperture) —

(i) Lens: focused light onto the image sensor

(ii) Aperture (f-stop): controls the amount of ^{incoming} light

(iii) Smaller aperture $\rightarrow$ Greater depth of field, less light

(iv) Lens distortion may curve straight lines

(v) Focal length determines the field of view (FOV)

Image Sensor (CCD & CMOS) -

- (i) Sensor converts light into electrical signals
- (ii) CCD: Better image quality, slower, higher power, common
- (iii) CMOS: Faster, cheaper, lower power, widely used
- (iv) Each pixel records light intensity
- (v) Resolution depends on the total number of pixels

Analog-to-Digital Conversion (ADC) -

- (i) Converts analog sensor output into digital values
- (ii) Sampling: captures signal at discrete time intervals
- (iii) Quantization: converts continuous values into discrete level
- (iv) Bit Depth: determines intensity levels
 $8 \text{ bits} = 256 \text{ levels (0-255)}$
- (v) Higher sampling rate $\rightarrow$ Better temporal accuracy

Image Size Calculation -

Formula:

- Pixels = width $\times$ height

- Bits = Pixels $\times$ bit depth

- Bytes = Bits / 8

Sensor Selection Pitfalls -

- (i) Poor low-light performance $\rightarrow$ Noisy images
- (ii) Lens and sensor resolution mismatch $\rightarrow$ Loss of detail
- (iii) Low frame rate $\rightarrow$ motion blur
- (iv) Poor dynamic range $\rightarrow$ Loss of shadow details
- (v) Wrong interface $\rightarrow$ Bandwidth limitations

Lighting & Illumination -

- (i) Proper illumination is crucial for image quality
- (ii) Direct lighting: enhances texture, creates shadows
- (iii) Diffuse lighting: uniform illumination, reduces glare
- (iv) Backlighting: best for edge detection & dimension measurement
- (v) Structured lighting: used for 3D shape reconstruction

Data Processing & Interfaced -

- (i) Common interfaces: USB, Camera Link, GigE
- (ii) FPGAs: Accelerates real time image processing
- (iii) Standard protocols ensure device compatibility
- (iv) Data compression reduces transmission bandwidth
- (v) Synchronization ensures multiple cameras capture simultaneously

Camera as Optical System -

- (i) Camera projects a 3D scene onto a 2D sensor
- (ii) lens focuses light onto the sensor
- (iii) Aperture controls light intake and depth of field
- (iv) Shutter speed controls exposure time
- (v) ISO controls sensitivity (noise)
- (vi) Focal length determines Field of View (FOV)
- (vii) Pinhole camera model is the basic image formation model

Lens Types & Focal length -

- (i) Prime (fixed) lens: Sharper image, fixed FOV
- (ii) Zoom lens: variable focal length
- (iii) Wide angle: large FOV, barrel distortion
- (iv) Telephoto: Narrow FOV, magnified distant objects
- (v) Fish-eye: Ultra-wide ($>180^\circ$) FOV
- (vi) Distortions: Barrel (outward) & Pin cushion (inward)

Specialized Camera Systems -

Industrial Vision:	Medical & Scientific:
- Machine vision	- Endoscopic cameras
- Line scan cameras	- Microscope cameras
- High speed cameras	- Hyperspectrum cameras
- Stereo vision	- Confocal microscopy
- Multispectral cameras	- OCT (Optical Coherence Tomography)

Field of View (FOV) -

Formula: $FOV = 2 \times \arctan\left(\frac{\text{Sensor width}}{2 \times \text{focal length}}\right)$

- (a) FOV depends on sensor size & focal length
- (b) Larger focal length $\rightarrow$ smaller FOV
- (c) Larger sensor $\rightarrow$ larger FOV

Camera Selection Pitfalls -

- (i) Wrong sensor size $\rightarrow$ incorrect FOV
- (ii) Lens-sensor mismatch
- (iii) Ignoring lighting
- (iv) Rolling shutter for fast motion
- (v) Ignoring dynamic range
- (vi) Not checking FPS/exposure

Camera Calibration -

- (i) Intrinsic parameters: focal length, principal point
- (ii) Extrinsic parameters: camera position & orientation
- (iii) Corrects radial & tangential distortion
- (iv) Checkboard $\rightarrow$ common method
- (v) Essential for 3D reconstruction & accurate measurements

Emerging Camera Technologies -

- (i) Event based cameras
- (ii) Neuromorphic sensors
- (iii) Computational photography
- (iv) Solid state LiDAR
- (v) AI cameras

Pinhole Camera Model Definition -

- (i) Assumes light travels in straight lines through a single point (optical center)
- (ii) Assumes very small aperture
- (iii) Sees idealized geometric world

Perspective Projection Equations -

$$x = f \times \left(\frac{x}{z} \right) \quad f = \text{focal length}$$
$$z = \text{depth (3rd dimension)}$$

Similarly for vertical coordinates, $y = f \times \left(\frac{y}{z} \right)$

Role of Focal length -

- (i) Determines magnification of projected image
- (ii) Longer focal length $\rightarrow$ larger image
- (iii) Short focal length $\rightarrow$ wide FOV but less magnification
- (iv) Long focal length $\rightarrow$ narrow FOV but high magnification

From Image to Pixel Coordinates -

Horizontal coordinates, $u = (x/S_x) + C_x$

$C_y/C_x = \text{principle point}$

Vertical coordinates, $v = (y/S_y) + C_y$

$S_x/S_y = \text{scaling factors}$

$\rightarrow$ Pixel location of optical center

Common Pitfalls in Projection -

- (i) Negative z values are not visible
- (ii) Lens distortion causes straight lines to appear curved
- (iii) Ignoring principle point offset leads to systematic errors
- (iv) Real world lighting can further complicate the projection process

Matrix Representation of Projection -

$$[u, v, 1]^T = P \times [X, Y, Z, 1]^T$$

Camera Calibration -

- (i) Estimates parameters to correct lens distortions & projection errors
- (ii) Maps 3D world coordinates to 2D image pixel coordinates

Intrinsic Parameters -

Describes internal geometry of the camera itself

- Focal length (magnification & FOV)
- Principle point (optical center)
- Radial & tangential distortion (corrects lens shape imperfections)
- Skew coefficients (non orthogonal pixel axes)

Extrinsic Parameters -

Define camera's position & orientation

- Rotation matrix (camera's angular orientation)
- Translation vector (camera location in 3D space)

Calibration Patterns -

Provides 3D points for the algorithm to match

- Checkboard patterns (easy corner detection)
- Circular grid patterns (when lighting conditions are challenging)

Required: Physical size of the pattern elements

Multiple views of the pattern

Equation: $x = f_x \frac{X}{Z} + c_x$ $y = f_y \frac{Y}{Z} + c_y$ $Z = \text{depth of the point from camera}$

$$y = f_y \frac{Y}{Z} + c_y$$

f_x, f_y = focal length expressed in pixels
 c_x, c_y = principal point

* Good calibration typically achieves an error of less than 0.5 pixels

eg. Given a point at $X=100\text{mm}$, $Y=50\text{mm}$, $Z=1000\text{mm}$ in camera frame

Assume focal length $f_x=1000$ & principle point $c_x=500$

$$x = (1000 \times 100) + 500$$

$$= 600 \text{ pixels}$$

Optimization process -

- Minimizes reprojection error between observed and predicted points
- Reprojection error = distance between detected - corners and model predictions

- Goal: Find parameters that make the model fit the data best

Common Pitfalls -

- Using too few images
- Poor lighting or low contrast
- Ignoring lens distortion
- Calibrating with a pattern that is not perfectly flat

Radiometry vs Photometry -

Aspect	Radiometry	Photometry
1) Measures	Absolute physical radiation energy	Radiation weighted by human eye sensitivity
2) Based on	Physical energy	Human visual perception
3) Units	Watts (W)	Lumen (lm)
4) Spectrum	Visible & non visible	Mainly visible
5) Image Formation	Used for accurate physical image formation	Can fail for non visible applications
		electromagnetic radiation

Radiant Flux & Intensity -

- Radiant flux (Φ) is the total power emitted by the source. Represents rate of energy transfer of
- Radiant intensity (I) is flux per unit solid angle.
- Intensity describes how directional a light source is
- Unit for flux is Watts, unit for intensity is Watts per steradian

Irradiance & Image formation -

- Irradiance (E) is the power incident on a surface area
- Image formation depends on irradiance
- Irradiance determines signal strength captured by pixels

- Unit = Watts per square meter
- Higher irradiance results in brighter pixel values

Lambertian Surface Model -

- Surface that reflects light equally in all directions
- Radiance is constant
- Most matte surfaces approximate Lambertian behavior well
- Reflected radiance depends on surface albedo

fraction of light reflected by surface

Inverse square law -

- Light intensity decreases with the square of distance
- Irradiance is proportional to $\frac{1}{(\text{distance})^2}$
- Doubling distance reduces irradiance to one quarter
- Law applies to point sources in free space

Irradiance calculation -

Formula: $E = \frac{\Phi}{(4\pi d^2)}$ for isotropic point source

Radiance & Sensation -

- Radiance is power per unit area per unit solid angle
- Radiance is conserved along a ray in vacuum
- Does not change with distance
- Determines brightness of an image patch

Sensor response & exposure -

- Sensor response is linear to incident irradiance initially
- Saturation occurs when irradiance exceeds sensor capacity
- Exposure time controls total energy collected by pixels
- Aperture size controls rate of light collection

Nature of light in space -

- Light is electromagnetic radiation traveling as transverse waves through space
- Light travels in straight lines called rays
- Speed of light = 3×10^8 m/s
- Visible light spectrum range: 380 nm - 750 nm
- Frequency $\propto \frac{1}{\text{wavelength}}$

Luminous Flux -

- Luminous flux (Φ_v) measures light power weighted by human eye sensitivity
 - Unit: lumen (lm)
 - Accounts for visible spectrum perception
 - Photometry
 - Used for lighting design
- Appear as light spots (specular highlights)

Reflection Types -

- (i) Specular reflection: occurs on smooth surfaces like mirrors or polished metal. $\theta_i = \theta_r$
- (ii) Diffuse reflection: occurs on rough surfaces scattering light in all directions. Allows us to see objects from any viewing angle
- (iii) Lambertian surface: reflect light equally in all directions regardless of angle

Absorption and Transmission -

- Absorption occurs when a surface converts light energy into heat
- Absorbed fraction depends on the material properties & wavelength
- Transmission occurs when light passes through a transparent or translucent medium

- Opaque objects absorb or reflect all incident light without transmission

- Black surface absorb most visible light

Illumination Sources in Vision Systems -

- Provide energy required to form an image on the sensor

- Natural light varies in intensity & spectral composition

- Spectral power distribution determines how different materials reflect light wavelengths

- Geometric shadows occur when an opaque

object blocks path of light rays

- Umbra is the region of total shadow

- Penumbra is the region of partial shadow

- Cast shadows are projected onto other surfaces

- Attached shadows are self occlusions

- Size of light source determines sharpness of shadow edge

- Shadows create abrupt intensity drops while shading creates smooth intensity gradients

Intensity variation on surfaces -

- Shading refers to variation in pixel intensity due to surface orientation

- Shading does not involve occlusion

- Shading gradient reveals curvature & depth

- Intensity gradient reveals continuous & smooth

- Shading effects are continuous & smooth

Lambertian Reflectance Model -

$$I = L \times \cos \theta \quad \left| \begin{array}{l} L = \text{light intensity} \\ \theta = \text{angle} \end{array} \right.$$

- Intensity of a pixel $\propto$ cosine of angle of incidence

- Works well for matte surfaces

Impact on object detection & segmentation -

- Shadows can cause object detectors to miss targets

- Shading variations can be mistaken for texture changes

- Feature extraction techniques must focus on edges & shapes rather than edges

Triangulation calculates 3D position of the point in space

High level vision -

- Interprets the meaning of visual data semantically

- Transforms low level features into abstract concepts & labels

- Tasks:

(i) Image segmentation

(ii) Object recognition

(iii) Scene understanding

(iv) Action recognition

Role of geometric methods -

- Provide spatial measurements for 3D interpretation

- Relate 2D image coordinates to real world 3D object positions

- Stereo vision uses 2 views to reconstruct depth & geometry

From pixels to semantics -

(i) Starts with raw pixel intensity values

(ii) Low level filters extract edges, corners, etc

(iii) Feature descriptors encode these local patterns into numerical vectors

Disparity is horizontal shift of a point between left & right images

(iv) Classifier on neural networks map features to semantic categories

Depth, $Z = \frac{f \times B}{d}$ | B = baseline: is the distance between the 2 camera centers

Applications of high level vision -

(a) Autonomous vehicles

(b) Medical imaging

(c) Surveillance systems

(d) Augmented reality

(e) Robotics

Semantic segmentation techniques -

- Assigns a class label to every pixel in an image

(a) Full Convolutional Networks (FCN)

(b) U-Net architecture

- Output is a dense map where each pixel has a specific meaning

Homography Transformation -

- Homography maps points between 2 images of a planar surface

- Handles perspective distortions | - It is a 3×3 matrix that relates corresponding points on a flat plane

Model Based Vision -

- Uses explicit 3D geometric representations of objects
- Relies on prior knowledge of the target object's shape
- Reduces search space by constraining possible object locations
- Particularly effective for known objects in structured environments

3D model representation -

- Models are typically represented as polygon meshes or boundary representations
- Surface normals provide orientation info for lighting & reflection
- Models must be scaled & oriented relative to the camera
- Geometric invariants like angles & ratios remain constant under rotation
- 2D features are projected onto the 3D model surface
- Optimization algo minimize error between projected & observed features

Pose estimation & alignment -

- Determines object's position & orientation in 3D space
- Transformation matrix maps model coordinates
- Rotation & translation parameters are solved using geometric constraints
- Multiple views can resolve ambiguities
- Least squares methods are commonly used to find the optimal pose

Advantages -

- (i) High accuracy in detection
- (ii) Robustness to partial occlusion
- (iii) Ability to estimate 3D pose directly from a single image
- (iv) Works well in controlled environments with known lighting conditions

Challenges & Limitations -

- (i) Computationally expensive
- (ii) Sensitive to lighting changes
- (iii) Struggles with significant deformations
- (iv) Initial pose estimation can be difficult

Pose Consistency -

- Measures how well multiple 3D pose estimates agree geometrically
- Quantifies the overlap between rotation & translation parameters of hypothesis
- High consistency $\Rightarrow$ multiple detections likely refer to same physical object
- Low consistency $\Rightarrow$ detections are outliers

Problem of multiple hypotheses -

- Single view detection system produces multiple overlapping bounding boxes for one object
- Different algorithms may generate conflicting pose estimates for the same target
- Noise & occlusion create false positives
- Geometric methods resolve this by grouping similar poses into a single cluster

Pose Clustering Algorithm -

- Groups pose ~~object~~ estimates based on similarity in rotation & translation
- Uses distance metric to determine if 2 ~~pos~~ poses belong to the same cluster

- Outliers receive less or no votes & are discarded
- Voting scheme assigns a weight to each pose
- Clusters with highest total weight are selected as the final object poses
- Poses iterates until all poses are either clustered or marked as outliers

Distance metric for Poses -

$$d = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2 + \alpha \cdot \Delta \theta^2}$$

- Pose distance combines translation error & rotation error into a single scalar
- Translation error is the Euclidean distance between 2 true 3D position vectors
- Rotation error is measured as the angle of the ~~min~~ minimal rotation
- Smaller thresholds create tighter clusters but risk splitting a single object
- Larger thresholds merge distinct objects that are close together

Geometric invariants -

- When an object's feature stays constant despite viewing changes
- Allow systems to recognise the same

- Object from different angles
- Reduce search space for matching objects

Types of Transformations -

- (i) Translation shifts the object's position without changing its shape or size
- (ii) Rotating turns the object around a point
- (iii) Scaling ^{changing} the size of the object
- (iv) Affine transformations preserve parallel lines & ratios of lengths
- (v) Projective transformations model perspective objects

Scale invariance & ratios -

- Enables recognition regardless of the object's distance from the camera
- Ratios of lengths are often invariant
- Area ratios between sub regions also remain constant
- Ratio of a rectangle's width to its height remains constant

Rotation invariance & angles -

- Allows recognition when an object is turned at any angle
- Internal angles remain constant
- Dot products of vectors are invariant
- Angle between 2 edges of a Δ remains constant

Affine invariants & ratios -

- Ratios of lengths along parallel lines are preserved under affine transformations
- Ratios of areas of 2 shapes is invariant
- Cross ratios of 4 collinear points are invariant

$$\text{Aspect Ratio} = \frac{\text{Width}}{\text{Height}}$$

Verification in Vision -

- Confirms if a geometric model fits the observed image data correctly
- Checks if estimated parameters satisfy the underlying geometric constraints
- Validates consistency of a hypothesized structure

Reprojection error as a metric -

- Measures distance between observed & predicted image points
- Calculated as euclidean distance
- Low error indicates a good fit between the model & data
- High error suggest outliers

Hypothesized line equation: $y=x$

RANSAC for robust verification -

- Iteratively selects random subsets to fit geometric models
- Verifies model by counting inliers within a distance threshold
- Outliers are excluded
- Repeats until a model with sufficient inliers is found

Epipolar Geometry verification -

$$x'^T F x = 0$$

- A point in one image must lie on the epipolar line

- Verification checks if the correspondence satisfies the epipolar constraint

Statistical test for verification -

- (i) χ^2 -squared test = gaussian distribution
- (ii) Student's t test = mean errors
- ~~(iii)~~ - Provide probabilistic confidence in the verification result

3D reconstruction verification -

- Features reconstructed 3D points are consistent
- Triangulation errors are checked
- Inconsistent 3D points indicate errors in feature matching
- Bundle adjustment refines parameters to minimize reprojection errors

Applications of High level vision -

- (i) Scene understanding (categorizes the entire environment)
 - Autonomous navigation
 - Surveillance
- (ii) 3D reconstruction from 2D images
 - Robotics
 - Augmented Reality

(iii) Autonomous vehicles (use geometric methods for object detection, 3D reconstruction & visual odometry)

(iv) Augmented reality (use geometric tracking, scene understanding, depth estimation)

- Gaming
- Education
- Industrial maintenance